

Regression on Acoustic Emotion Data: Predicting Arousal and Valence

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Abstract—This project uses the EmoSounds and IADSED acoustic emotion datasets to predict arousal and valence from numerical audio features. The analysis first cleaned the data, removed non-predictive metadata, checked for constant features, saved preprocessed CSV files, and held out a final 20% test set. The remaining 80% was used for 5-fold cross validation and hyperparameter tuning for linear regression, decision tree regression, and random forest regression. We also repeated the complete modeling procedure after selecting the 20 features with the largest F-regression scores. Random forest regression produced the lowest held-out test RMSE for all four dataset-target combinations when all 68 acoustic features were used: 0.2106 and 0.3828 for EmoSounds arousal and valence, and 0.7138 and 1.1301 for IADSED arousal and valence. Feature selection reduced the decision tree test error for both IADSED targets, but it did not improve the best random forest result. The results show that nonlinear ensemble models were the strongest predictors in this experiment, although the gap between training and test error also shows that the random forests overfit more than the simpler models.

Index Terms—acoustic emotion recognition, regression, arousal, valence, cross validation, feature selection, random forest

I. INTRODUCTION

The main goal of this project was to predict emotional dimensions from acoustic features using two datasets, EmoSounds and IADSED. The two target variables were arousal and valence. We used the same general modeling procedure for both datasets so that the results could be compared in a consistent way. The project required data cleaning, visualization, preprocessing, a 60:20:20 train-validation-test split, 5-fold cross validation on the combined train and validation data, hyperparameter tuning, three regression models with different levels of complexity, and feature selection.

The analysis focused on three questions. First, we wanted to know how well a simple linear model could predict the targets from the acoustic variables. Second, we wanted to see whether nonlinear decision tree and random forest models could improve the prediction error. Third, we wanted to know whether reducing the input to the top 20 F-regression features would improve generalization or mainly remove useful information. We used root mean squared error (RMSE) throughout the project because the targets are continuous and lower RMSE directly represents smaller prediction error.

II. DATASET DESCRIPTION AND CLEANING

The raw EmoSounds file contained 600 rows and 75 columns, while the raw IADSED file contained 935 rows and 76 columns.

TABLE I: Dataset cleaning and split summary.

Item	EmoSounds	IADSED
Raw rows	600	935
Raw columns	75	76
Missing cells	0	27
Rows removed for missing values	0	8
Duplicate rows found	0	0
Clean rows	600	927
Acoustic predictor features	68	68
Train rows (60%)	360	555
Validation rows (20%)	120	186
Test rows (20%)	120	186
Train+validation rows (80%)	480	741

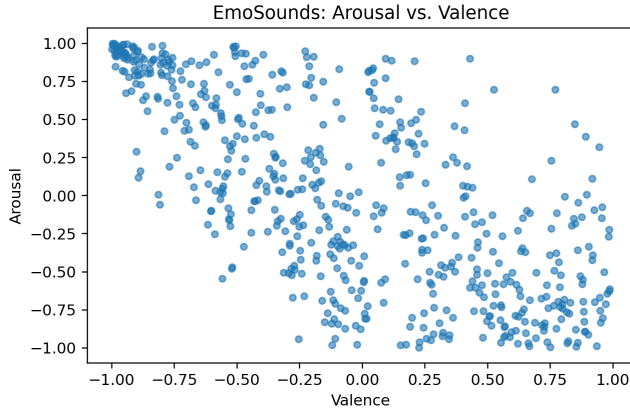
EmoSounds had no missing values and no duplicated rows. IADSED had 27 missing cells distributed across 8 rows and no duplicated rows. As a cleaning technique, duplicate rows were removed from both datasets, and the 8 IADSED rows containing missing values were dropped. This left 600 EmoSounds observations and 927 IADSED observations.

The targets use different numerical scales in the two datasets. EmoSounds arousal and valence are approximately on a $[-1, 1]$ scale, while IADSED values are on a larger positive scale. Because of this, RMSE values should be compared across models within the same dataset and target rather than directly between EmoSounds and IADSED.

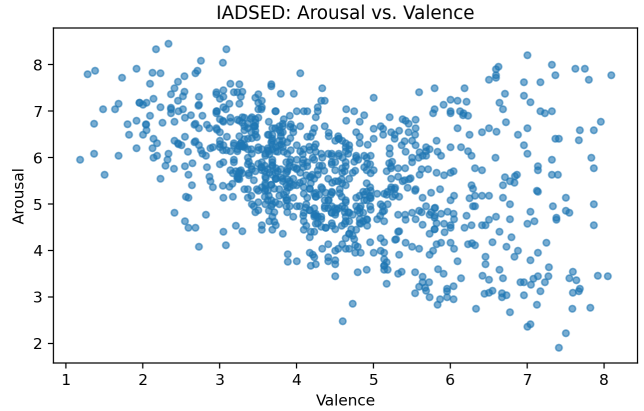
III. VISUALIZATION AND PREPROCESSING

The first visualization compared arousal and valence directly. As shown in Fig. 1, both datasets have a negative relationship between the two targets. The relationship is much stronger in EmoSounds, where the Pearson correlation after cleaning is approximately -0.711 , compared with approximately -0.399 in IADSED. This means that higher valence is generally associated with lower arousal in both datasets, but the pattern is more concentrated in EmoSounds.

For preprocessing, columns that represented file names, text labels, dataset identifiers, categories, genres, and other metadata were removed because the models were intended to use acoustic features. The dominance target was also removed from IADSED because it was not required for this project. We then kept numerical columns and checked for constant variables. No constant acoustic variables remained in either preprocessed dataset, so both final files contained 68 predictor features plus the two targets. The resulting datasets were saved as `EmoSounds_preprocessed.csv` and `IADSED_preprocessed.csv`.



(a) EmoSounds.



(b) IADSED.

Fig. 1: Arousal versus valence after data cleaning. EmoSounds has a visibly stronger negative relationship between the targets than IADSED.

Scaling was handled inside the modeling pipeline. StandardScaler was applied before linear regression, while the decision tree and random forest did not require scaling. Keeping scaling inside the pipeline was important because the scaler was fit only from training data during each cross-validation fold rather than from the entire dataset.

IV. DATA SPLITTING, CROSS VALIDATION, AND TUNING

The test set was held out once before any model tuning. We first split each preprocessed dataset into 80% train+validation data and 20% test data. The 80% pool was also split into 60% train and 20% validation subsets to verify the required 60:20:20 proportions shown in Table I. The separate validation subset was not used as a second final evaluation set. Instead, the complete 80% train+validation pool was passed to 5-fold cross validation for hyperparameter selection, following the required project procedure.

GridSearchCV used shuffled 5-fold KFold cross validation with random seed 42 and selected the hyperparameter combination with the lowest average validation RMSE. After the best parameters were selected, GridSearchCV refit that configuration once on all train+validation rows. We then measured RMSE on the same 80% training pool and once on the untouched 20% test set. This keeps the test set separate from model selection.

The tuned linear regression parameter was `fit_intercept` $\in$ {True, False}. For the decision tree, the grid used `max_depth` $\in$ {5, None} and `min_samples_leaf` $\in$ {1, 3}. The random forest used 50 trees with the same depth and minimum-leaf grids. These grids were evaluated separately for each dataset, each target, and each feature setting.

V. PREDICTION MODELS AND FEATURE SELECTION

We compared three regression models with increasing flexibility. Linear regression was the simplest model and provided a baseline that assumes an approximately linear relationship between the acoustic features and each target.

A decision tree allowed nonlinear splits and interactions but could overfit if it became too deep. Random forest regression combined 50 decision trees and averaged their predictions, making it the most flexible model used in this project.

For feature selection, SelectKBest with the F-regression score selected the 20 acoustic variables most strongly associated with the current target. The selector was placed at the beginning of the pipeline and therefore refit inside every cross-validation fold. This detail prevents information from the held-out fold from leaking into feature selection. The entire tuning and evaluation procedure was then repeated using the selected features so that performance before and after feature selection could be compared directly.

The selected sets differed by dataset and target, but common variables included RMS dynamics, spectral centroid/spread/flatness, rolloff measures, MFCC statistics, roughness, and timbre spectral flux. This suggests that both energy-related and spectral/timbre information contributed to the target relationships rather than one single acoustic family being sufficient by itself.

A. Evaluation Metric

The project used RMSE for cross validation and final evaluation. For n observations, true values y_i , and predictions $\hat{y}_i$, RMSE is

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}. \quad (1)$$

Lower values are better. Training RMSE was computed on the final train+validation pool after refitting, and test RMSE was computed once on the held-out test set.

VI. RESULTS

Table II reports cross-validation, training, and test RMSE for every dataset-target-model combination before and after

feature selection. Figure 2 provides a visual comparison of the held-out test results.

A. Best Hyperparameters

Table III lists the final parameters selected by cross validation. For compactness, “d” is maximum tree depth, “l” is minimum samples per leaf, and ∞ indicates `max_depth=None`. The random forest used 50 trees in every row.

VII. ANALYSIS AND DISCUSSION

The most consistent result is that random forest regression gave the lowest held-out test RMSE in all four prediction tasks when the full feature set was used. For EmoSounds, the random forest test RMSE was 0.2106 for arousal and 0.3828 for valence. For IADSED, the corresponding values were 0.7138 and 1.1301. This suggests that nonlinear relationships and interactions between acoustic features are important for these targets. The linear model was competitive in some cases, especially EmoSounds arousal, but it was not the best final model for any of the four tasks.

Feature selection did not have one uniform effect. On EmoSounds, reducing the inputs to 20 features increased test RMSE for all three models on both targets. This suggests that information useful for prediction was distributed across more than the 20 features chosen by the univariate F score. On IADSED, feature selection improved the decision tree test RMSE from 0.9040 to 0.8731 for arousal and from 1.2610 to 1.2145 for valence. However, it still did not beat the full-feature random forest. The feature-selection comparison therefore shows that a smaller feature set can help some lower-capacity models, but the best overall results in this experiment used all 68 acoustic predictors.

There is also clear evidence of overfitting in the random forest models. For example, the full-feature IADSED valence random forest had a training RMSE of 0.5153 but a test RMSE of 1.1301. The full-feature EmoSounds valence random forest had a training RMSE of 0.1389 and a test RMSE of 0.3828. The decision trees also had lower training error than test error, but the gaps were generally smaller. Linear regression had the smallest train-test gaps because it has much lower flexibility. That being said, the random forest still had the lowest test RMSE, so some overfitting was present without removing its advantage over the simpler models.

The cross-validation results were generally consistent with the held-out test results for selecting the random forest as the strongest model. There were still differences between CV and test rankings for some linear/tree comparisons, which is expected because the datasets are not very large and each split contains a finite sample of sound events. This is one reason the final conclusions are based on the untouched test set rather than only the training error.

VIII. BENCHMARK NOTE AND REPRODUCIBILITY

The supplied assignment lists benchmark fields for Accuracy and F1-score as “TBD.” Those metrics are also classification metrics, while this project is a regression task evaluated with

RMSE. Therefore, there was no numeric benchmark value in the supplied project materials that could be matched or exceeded. The reproducible comparison in this report is based on the required RMSE values for train+validation and test data.

The notebook uses random seed 42 for splitting, cross-validation shuffling, the decision tree, and the random forest. The provided notebook was executed from beginning to end using the supplied raw CSV files, and it completed without execution errors. The preprocessed CSV files regenerated by that run contained 600 rows by 70 columns for EmoSounds and 927 rows by 70 columns for IADSED.

IX. CONCLUSION AND FUTURE WORK

This project compared linear regression, decision tree regression, and random forest regression for predicting arousal and valence from acoustic features in EmoSounds and IADSED. We cleaned the datasets, saved preprocessed numeric files, verified the required 60:20:20 split, tuned each model using 5-fold cross validation on the 80% train+validation pool, retrained the selected configuration on that full pool, and evaluated it once on the final test set. We also repeated the complete modeling process using the top 20 F-regression features.

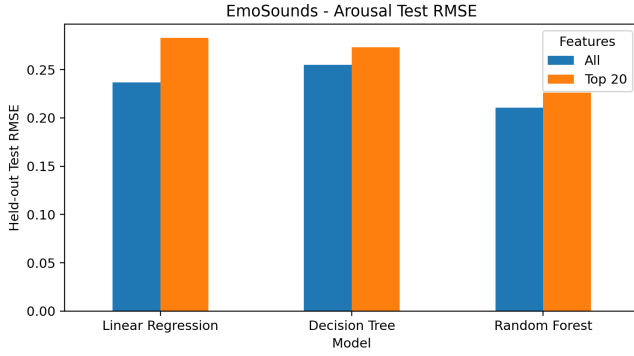
The main conclusion is that random forest regression was the strongest model across all four prediction tasks, although it also showed the largest training-to-test gaps. Feature selection was useful for the IADSED decision tree but did not improve the best overall test score. These results suggest that the emotional targets depend on nonlinear combinations of multiple acoustic variables and that aggressive univariate feature reduction can remove useful information. In future work, the modeling could be improved by testing a wider random forest grid, adding regularized linear models or gradient boosting, and comparing feature-selection methods that account for interactions rather than ranking each feature independently.

PROJECT LINK

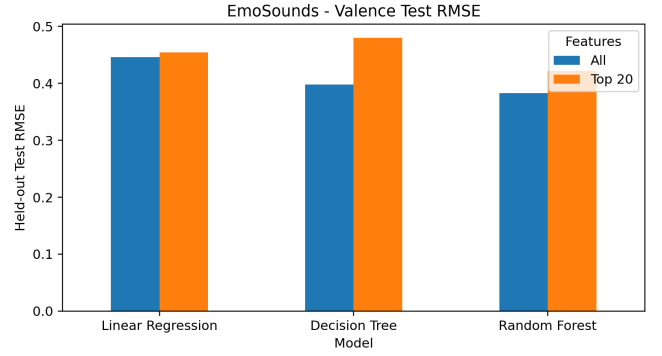
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TABLE II: RMSE before and after top-20 F-regression feature selection. “CV” is the best mean 5-fold cross-validation RMSE on the train+validation pool.

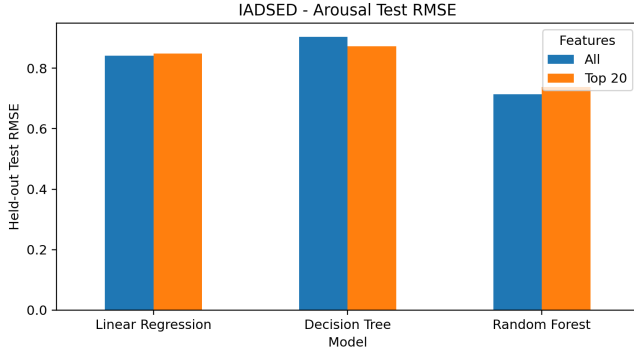
Dataset	Target	Model	CV All	Train All	Test All	CV Top20	Train Top20	Test Top20
EmoSounds	Arousal	Linear	0.3335	0.2441	0.2366	0.3435	0.2918	0.2828
EmoSounds	Arousal	Tree	0.3273	0.1763	0.2550	0.2926	0.1926	0.2729
EmoSounds	Arousal	Forest	0.2461	0.0902	0.2106	0.2499	0.0904	0.2258
EmoSounds	Valence	Linear	0.3903	0.3262	0.4462	0.4094	0.3929	0.4545
EmoSounds	Valence	Tree	0.4401	0.2784	0.3982	0.4299	0.3109	0.4799
EmoSounds	Valence	Forest	0.3614	0.1389	0.3828	0.3739	0.1912	0.4220
IADSED	Arousal	Linear	0.9169	0.7681	0.8422	0.8626	0.8404	0.8494
IADSED	Arousal	Tree	0.9795	0.7024	0.9040	0.8938	0.7162	0.8731
IADSED	Arousal	Forest	0.8058	0.3133	0.7138	0.7914	0.3019	0.7380
IADSED	Valence	Linear	1.2039	1.0698	1.2128	1.2058	1.1546	1.2396
IADSED	Valence	Tree	1.2776	0.9550	1.2610	1.2238	0.9811	1.2145
IADSED	Valence	Forest	1.1214	0.5153	1.1301	1.1503	0.5539	1.1438



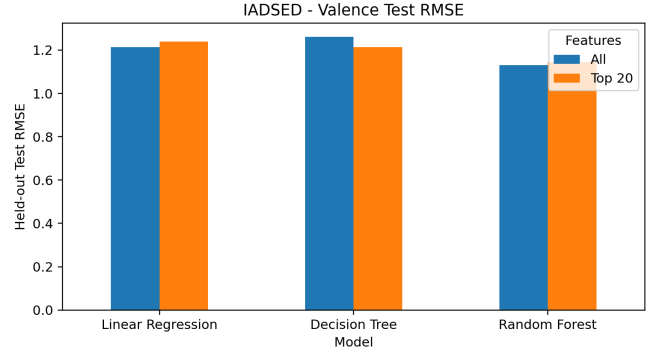
(a) EmoSounds arousal.



(b) EmoSounds valence.



(c) IADSED arousal.



(d) IADSED valence.

Fig. 2: Held-out test RMSE with all 68 features and with the top 20 F-regression features. Lower is better.

TABLE III: Cross-validation-selected hyperparameters before and after feature selection.

Dataset	Target	Model	All Features	Top 20 Features
EmoSounds	Arousal	Linear	intercept=True	intercept=False
EmoSounds	Arousal	Tree	d=5, l=1	d=5, l=3
EmoSounds	Arousal	Forest	d=∞, l=1	d=∞, l=1
EmoSounds	Valence	Linear	intercept=True	intercept=True
EmoSounds	Valence	Tree	d=5, l=3	d=5, l=3
EmoSounds	Valence	Forest	d=∞, l=1	d=∞, l=3
IADSED	Arousal	Linear	intercept=True	intercept=True
IADSED	Arousal	Tree	d=5, l=3	d=5, l=3
IADSED	Arousal	Forest	d=∞, l=1	d=∞, l=1
IADSED	Valence	Linear	intercept=True	intercept=True
IADSED	Valence	Tree	d=5, l=1	d=5, l=3
IADSED	Valence	Forest	d=∞, l=3	d=∞, l=3